Code No: 124CN **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year IISemester Examinations,August/September - 2022
COMPUTER ORGANIZATION
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1. What is addressing mode? List and explain the different addressing modes. [15]

2.a) Discuss conditional branch instructions with examples.
b) What are status bits? Explain with block diagram of 8-bit ALU with a 4-bit status register. [7+8]

3. What is Asynchronous data transfer? Explain different data transfer methods. [15]

4.a) Explain DMA controller with the block diagram.
b) Explain IOP of a computer with two processors. [8+7]

5. Describe the design of a 1024 × 8 RAM with the help of 128 × 8 RAM chips and 512 × 8 ROM chips. [15]

6. Define Mapping. Explain briefly the mapping procedures used in organization of cache memory. [15]

7. State and explain the different instruction formats of 8086. [15]

8.a) Draw and discuss the CPU pin diagram of 8086.
b) Draw and discuss the Flag register of 8086 in brief. [8+7]

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